

ARNOLD MACWAN

AI-ML Engineer | Software Engineer | Passionate Tech Learner

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SUMMARY

Proficient software developer and AI/ML Engineer with hands-on experience building and deploying machine learning models, LLM-powered applications, and full-stack systems (MERN, FastAPI, Python). Skilled in the complete data science lifecycle: data preprocessing, exploratory analysis, model development, evaluation, and production deployment via REST APIs. Delivered 6+ end-to-end AI and web applications spanning predictive modeling, NLP/RAG pipelines, and computer vision. Combines strong analytical and research skills with proven communication ability, demonstrated through technical training delivered to the students as Visiting Faculty and a published, internationally presented research paper.

EXPERIENCE

AI-ML Trainee — Hyperlink Infosystem Pvt. Ltd.

Dec 2025 – July 2026

- Built and evaluated **machine learning models** for real-world problem statements using Python, Scikit-learn, and TensorFlow, taking each from raw data to a validated, deployable solution.
- Engineered end-to-end AI pipelines covering data preprocessing, feature engineering, model training, evaluation, and deployment, reducing manual handoff between pipeline stages.
- Designed and shipped **production-ready REST APIs** using FastAPI to serve ML models, enabling real-time inference for downstream applications. Integrated Large Language Models (LLMs) into automation workflows for text generation and processing, cutting manual content-handling effort for the team.

Visiting Faculty — LJSCA

Jul 2024 – Oct 2025

- Delivered hands-on technical training in Django and ASP.NET MVC to the students across multiple cohorts, combining live coding with project-based learning.
- Designed structured lesson plans and assessments that improved student project completion and hands-on coding proficiency. Strengthened technical presentation and public speaking skills through continuous student feedback and iterative content refinement.

PROJECTS

Stock Analysis & Forecasting | Python, Jupyter, Scikit-learn

- Performed end-to-end exploratory data analysis (EDA) on multi-asset market datasets sourced from Kaggle and yfinance, identifying key trend and volatility indicators. Built a machine learning prediction model for crypto market trend forecasting, applying data preprocessing and feature engineering to improve signal quality.

Live Employee Tracker | Python, Yolo, Reactjs

- Live video stream-based employee tracking system that detects and tracks individuals in real time using computer vision and deep learning. It enables monitoring of presence, movement, and activity patterns to improve workplace efficiency, security, and data-driven decision-making.

CognitiveWizard AI — Adaptive Learning Platform | FastAPI, React, LLMs, RAG, Langchain

- Built a scalable, microservices-ready backend using Python and FastAPI with asynchronous task processing for high-performance AI services.

- Built a multi-source content pipeline generating LLM-based summaries from PDFs, URLs, and YouTube content, plus a RAG-powered chatbot for interactive knowledge retrieval.
- Implemented NLP-based answer evaluation, semantic scoring, and sentiment analysis to drive personalized, adaptive learning pathways.

TaskQuest — Task Management System | MERN Stack

- Developed a multi-role task management platform with role-based access control and secure authentication. Built a real-time collaborative task board with live updates and a comment system, improving team communication and task visibility.
- Designed an admin dashboard for tracking team performance metrics.

SkillSwap — Peer Skill Exchange Platform | MEAN Stack

- Built a community-driven platform connecting users to teach and learn skills from one another.
- Implemented role-based access control with OAuth verification and a real-time chat feature.
- Designed an interactive admin dashboard with live KPIs and usage visuals for platform monitoring.

Primary HealthHub — Healthcare Management System | Django

- Built a role-based healthcare platform supporting Patients, Staff, and Admin with appointment booking and medical records management.
- Implemented strong validation and authentication alongside dynamic staff allocation and real-time administrative reporting.

EDUCATION

Data Science — TOPS Technologies2025 – 2026

Grade: B+

MSc (Information Technology) — LJSCA2024 – 2026

Grade: 7.6

SKILLS

Programming & Databases: Python, JavaScript, C, C++, MySQL, MongoDB, Vector DB

Data Science & Machine Learning: Data Analysis & Preprocessing, Machine Learning, Computer Vision, LLMs & RAG, Generative AI, AI Systems, Basic Agentic AI, Scikit-learn, TensorFlow, Langchain

Web Development: ReactJS, Angular, NodeJS, FastAPI, MVC Architecture, Microservices

Tools & Platforms: Git & GitHub, Docker, REST APIs, Postman, AI-Assisted Development, MS Office

Soft Skills: Quick Adaptability, Communication, Leadership, Presentation, Collaborative Work, Team Management

RESEARCH

“Metaverse Demystified: Design Insights, Deployment, Intricacies and Mitigations”

- Presented at ICRAECA 2025 (IFERP), an international conference held at LJ University.
- Explored core metaverse fundamentals, real-world deployment challenges, and proposed mitigation strategies.
- Published in the JSMS journal — DOI: 10.47974/JSMS-1537.